KAPPA-GENERATED EQUIVARIANT QUASI-COHERENT MODULES ON A GROUPOID SCHEME ARE SET-MANY UP TO ISOMORPHISM

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Allowed standard prerequisites (editorial).

- Lemma properties-lemma-set-of-iso-classes (canonical tag 077M). cited at proof lines 3364-3366. Isomorphism classes of kappa-generated quasi-coherent modules form a set.

Source reference. Original lemma 077M: sources/stacks-chow/tags.txt, source lines 9207–9207. Cited at proof lines 3364-3366: the isomorphism classes of kappa-generated quasi-coherent O_U-modules form a set; load-bearing.

Lemma 0.2. *Let (U, R, s, t, c) be a groupoid scheme over S. Let κ be a cardinal. There exists a set T and a family $(\mathcal{F}_t, \alpha_t)_{t \in T}$ of κ-generated quasi-coherent modules on (U, R, s, t, c) such that every κ-generated quasi-coherent module on (U, R, s, t, c) is isomorphic to one of the $(\mathcal{F}_t, \alpha_t)$.*

Proof. For each quasi-coherent module $\mathcal{F}$ on U there is a (possibly empty) set of maps $\alpha : t^*\mathcal{F} \to s^*\mathcal{F}$ such that $(\mathcal{F}, \alpha)$ is a quasi-coherent modules on (U, R, s, t, c). By Properties, Lemma 077M there exists a set of isomorphism classes of κ-generated quasi-coherent $\mathcal{O}_U$-modules. □

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